SHIBAJI CHAKRABORTY

Ph.D. | Research Scientist & Data Scientist

Space Weather · Ionospheric Physics · HF Propagation · ML/AI

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) — Ph.D. in Electrical Engineering (thesis) *08/2015 – 05/2021*

Virginia Polytechnic Institute and State University (Virginia Tech) — M.A. in Data Analytics and Applied Statistics *08/2019 – 12/2020*

Virginia Polytechnic Institute and State University (Virginia Tech) — M.S. in Electrical Engineering *08/2015 – 05/2017*

West Bengal University of Technology, Kolkata, India (WBUT) — B.Tech. in Electronics and Communication Engineering *08/2006 – 05/2010*

APPOINTMENTS

Embry-Riddle Aeronautical University — Research Scientist *09/2024 – Present*

Virginia Tech — Postdoctoral Associate *05/2021 – 09/2024*

Tata Consultancy Services — Information Technology Analyst *09/2010 – 07/2015*

RESEARCH GRANTS

Co-I (Institutional PI) of NASA LWS Research Award – "Ionospheric responses to thunderstorm-generated acoustic and gravity waves over the continental US" *2022 – 2026*

PI of NSF Award – "Collaborative Proposal: Investigation of ionospheric density response to American Solar Eclipses using coordinated radio observations with modeling support" *2024 – 2027*

Co-I of NASA Research Award – "Determining the connections between driving conditions, magnetosphere-ionosphere-ground properties, and extreme geomagnetic and geoelectric disturbances" *2024 – 2025*

PI of NSF-GEM Research Award – "Collaborative Research: GEM: Modeling Ionospheric and Magnetospheric Current Interactions with Submarine Cables" *2024 – 2027*

FELLOWSHIPS & AWARDS

Google — Google Academic Research Award *2025*

JSPS (Japan) — JSPS Postdoctoral Fellowship (Short-term) *2022*

Nagoya University — ISEE International Joint Research Fellowship *2022*

IEEE / WiSEE — Best Conference Paper *2021*

NCAR / UCAR — ASP-GVP Fellowship *2020*

Virginia Tech — Pratt Fellowship *2019*

LANL / ISR — Vella Fellowship and ISR-1 Summer School *2018*

Techno India — Won "Electronically Yours" event, category "Newron" *2010*

PROFESSIONAL SERVICES

AGU Fall Meeting — Primary Convener *2022*

AGU Fall Meeting — OSPA Judge *Since 2021*

NASA — Panel Reviewer of NASA Program *2023*

Virginia Tech — Mentor, Google Summer of Code *2018*

SuperDARN — Active member, SuperDARN Data Visualization Working Group *Since 2021*

Reviewer — Nature Communications; JGR: Space Physics; Space Weather; Radio Science; Astrophysical Journal; Earth and Space Science; IEEE Trans. Geoscience & Remote Sensing; Advances in Space Research; EGUsphere; Journal of Space Weather and Space Climate *Since 2019*

NSF — Panel Reviewer of NSF Program *2024*

CEDAR — Judge *2022 / 2024*

JOURNAL PUBLICATIONS

1. Chakraborty, S., Inchin, P. A., Debchoudhury, S., Bergsson, B., Zettergren, M., & Ruohoniemi, J. M. (2026). The Impacts of tropospheric gravity wave-generated MSTIDs on skywaves at mid-latitude north American sector observed and modeled using SuperDARN HF radars. Radio Science, 61, e2025RS008464. https://doi.org/10.1029/2025RS008464
2. Valentine, H. C., Barjatya, A., Chakraborty, S., Graves, N. P., Clayton, R. E., Debchoudhury, S., et al. (2026). Simultaneous multi-point in situ plasma density measurements of a mid-latitude sporadic E layer. Journal of Geophysical Research: Space Physics, 131, e2026JA035164. https://doi.org/10.1029/2026JA035164
3. Chakraborty, S., & Boteler, D. H. (2026). On the relationship between geomagnetically induced currents and geomagnetic field variations—Illustrated with case studies of pipeline and power systems in regions with different conductivity structures. Space Weather, 24, e2025SW004852. https://doi.org/10.1029/2025SW004852
4. Chakraborty, S., Hartinger, M. D., Shi, X., Boteler, D. H., Lawrence, E., Boteler, C., et al. (2026). Validating SCUBAS predictions of geomagnetically induced voltage in submarine cables using legacy superstorm observations. Space Weather, 24, e2025SW004843. https://doi.org/10.1029/2025SW004843
5. Bergsson, B., Inchin, P., Debchoudhury, S., Chakraborty, S., Snively, J., & Zettergren, M. (2026). Automated geolocation of diverse impulsive events using total electron content measurements. Frontiers in Astronomy and Space Sciences, 13–2026. doi:10.3389/fspas.2026.1812901
6. Chakraborty, S., Bullett, T., Barjatya, A., & Mabie, J. (2026). pynasonde: An open-source Python library for ionosonde data processing. SoftwareX, 34, 102617. https://doi.org/10.1016/j.softx.2026.102617
7. Wu, H., Wang, W., Shi, X., Lin, D., Hartinger, M. D., Baker, J. B. H., & Chakraborty, S. (2026). Global impacts of ultra-low-frequency waves: 1. Thermospheric responses and traveling atmospheric disturbances. Geophysical Research Letters, 53, e2025GL119835. https://doi.org/10.1029/2025GL119835
8. Chakraborty, S., Nishitani, N., Shi, X., Ponomarenko, P., Ruohoniemi, J. M., Baker, J. B. H., Coster, A. J., & Häggström, I. (2025). Solar Flare-Induced Gradient Drift Instability Observed by SuperDARN HF Radars. Journal of Geophysical Research: Space Physics, 130(10), e2025JA033824.
9. Bergsson, B., Debchoudhury, S., Inchin, P., Heale, C., Chakraborty, S., & Zettergren, M. (2025). The Impact of Receiver-Satellite Line-of-Sight Orientation on GNSS TEC Observations of TIDs Induced by Gravity Waves From Convective Sources. Journal of Geophysical Research: Space Physics, 130(9), e2025JA033938.
10. Inchin, P. A., Heale, C., Zettergren, M. D., Bergsson, B., Debchoudhury, S., & Chakraborty, S. (2025). Severe weather-generated acoustic and gravity wave impacts on the ionosphere: A model-guided case study. Journal of Geophysical Research: Space Physics, 130, e2025JA034012. https://doi.org/10.1029/2025JA034012

11. Prikryl, P., Themens, D. R., Chum, J., Chakraborty, S., Gillies, R. G., and Weygand, J. M. (2025). Observations of traveling ionospheric disturbances driven by gravity waves from sources in the upper and lower atmosphere. Ann. Geophys., 43, 511–534. https://doi.org/10.5194/angeo-43-511-2025

12. Chakraborty, S., Qian, L., Mrak, S., Mabie, J., Goncharenko, L., Mclnerney, J. M., & Bullett, T. (2025). Formation of the ionospheric G-condition following the 2017 Great American Eclipse. Earth and Space Science, 12, e2024EA004007. https://doi.org/10.1029/2024EA004007

13. Oberheide, J., Aggarwal, D., Bergsson, B., Chakraborty, S. et al. (2025). Impact of Terrestrial Weather on the Space Weather of the Ionosphere-Thermosphere: Initial Results from a NASA Living with a Star Focused Science Topic. Surv. Geophys. https://doi.org/10.1007/s10712-025-09895-7

14. Rout, D., Kumar, A., Singh, R., Patra, S., Karan, D. K., Chakraborty, S., et al. (2025). Evidence of unusually strong equatorial ionization anomaly at three local time sectors during the Mother's Day geomagnetic storm on 10–11 May 2024. Geophysical Research Letters, 52, e2024GL111269. https://doi.org/10.1029/2024GL111269

15. Shi, X., Chakraborty, S., Baker, J. B. H., Hartinger, M. D., Wang, W., Ruohoniemi, J. M., et al. (2025). Statistical characterization of joule heating associated with ionospheric ULF perturbations using SuperDARN data. Journal of Geophysical Research: Space Physics, 130, e2024JA033452. https://doi.org/10.1029/2024JA033452

16. Liu, X., Zhang, D., Coster, A. J., Xu, Z., Shi, X., & Chakraborty, S. (2024). The morphology and oscillations of nightside mid-latitude ionospheric trough at designated longitudes in the Northern Hemisphere. Journal of Geophysical Research: Space Physics, 129, e2024JA032864. https://doi.org/10.1029/2024JA032864

17. Inchin, P. A., Bhatt, A., Bramberger, M., Chakraborty, S., Debchoudhury, S., & Heale, C. (2024). Atmospheric and ionospheric responses to orographic gravity waves prior to the December 2022 cold air outbreak. Journal of Geophysical Research: Space Physics, 129, e2024JA032485. https://doi.org/10.1029/2024JA032485

18. Gowtam, V. S., Connor, H., Kunduri, B. S. R., Raeder, J., Laundal, K. M., Tulasi Ram, S., Chakraborty, S., et al. (2024). Calculating the high-latitude ionospheric electrodynamics using a machine learning-based field-aligned current model. Space Weather, 22, e2023SW003683. https://doi.org/10.1029/2023SW003683

19. Boteler, D. H., Chakraborty, S., Shi, X., Hartinger, M. D., & Wang, X. (2024). An examination of geomagnetic induction in submarine cables. Space Weather, 22, e2023SW003687. https://doi.org/10.1029/2023SW003687

20. Coyle, S. E., Baker, J. B. H., Chakraborty, S., Hartinger, M. D., Freeman, M. P., Clauer, C. R., et al. (2023). Substorms and solar eclipses: A mutual information-based study. Geophysical Research Letters, 50, e2023GL106432. https://doi.org/10.1029/2023GL106432

21. Reiss, M. A., Muglach, K., Mason, E., Davies, E. E., Chakraborty, S., Delouille, V., ... & Veronig, A. (2023). A Community Dataset for Comparing Automated Coronal Hole Detection Schemes. Astrophysical Journal Supplement. https://ore.exeter.ac.uk/repository/handle/10871/134745

22. Boteler, D., Chakraborty, S., Shi, X., Hartinger, M. and Wang, X. (2023). Transmission Line Modelling of Geomagnetic Induction in the Ocean/Earth Conductivity Structure. International Journal of Geosciences, 14, 767–791. doi: 10.4236/ijg.2023.148041

23. Rout, D., Patra, S., Kumar, S., Chakrabarty, D., Reeves, G. D., Stolle, C., Pandey, K., Chakraborty, S., Spencer, E. A. (2023). The growth of ring current/SYM-H under northward IMF Bz conditions present during the 21–22 January 2005 geomagnetic storm. Space Weather, 21, e2023SW003489. https://doi.org/10.1029/2023SW003489

24. Liu, J., Chakraborty, S., Chen, X., Wang, Z., He, F., Hu, Z., et al. (2023). Transient response of polar-cusp ionosphere to an interplanetary shock. Journal of Geophysical Research: Space Physics, 128, e2022JA030565. doi: 10.1029/2022JA030565

25. Collins, K., Gibbons, J., Frissell, N., Montare, A., Kazdan, D., Kalmbach, D., Swartz, D., Benedict, R., Romanek, V., Boedicker, R., Liles, W., Engelke, W., McGaw, D. G., Farmer, J., Mikitin, G., Hobart, J., Kavanagh, G., and Chakraborty, S. (2023). Crowdsourced Doppler measurements of time standard stations demonstrating ionospheric variability. Earth Syst. Sci. Data, 15, 1403–1418. doi:10.5194/essd-15-1403-2023

26. Chakraborty, S., Boteler, D. H., Shi, X., Murphy, B. S., Hartinger, M. D., Wang, X., Lucas, G., and Baker, J. B. H. (2022). Modeling geomagnetic induction in submarine cables. Front. Phys. 10:1022475. doi: 10.3389/fphy.2022.1022475

27. Prikryl, P., Gillies, R. G., Themens, D. R., Weygand, J. M., Thomas, E. G., and Chakraborty, S. (2022). Multi-instrument observations of polar cap patches and traveling ionospheric disturbances generated by solar wind Alfven waves coupling to the dayside magnetosphere. Ann. Geophys., 40, 619–639. doi: 10.5194/angeo-40-619-2022

28. Chakraborty, S., Qian, L., Baker, J. B. H., Ruohoniemi, J. M., Kuyeng, K., and Mclnerney, J. M. (2022). Driving influences of the Doppler flash observed by SuperDARN HF radars in response to solar flares. Journal of Geophysical Research: Space Physics, 127, e2022JA030342. doi: 10.1029/2022JA030342

29. Shi, X., Schmidt, M., Martin, C. J., Billett, D. D., Bland, E., Tholley, F. H., Frissell, N. A., Khanal, K., Coyle, S., Chakraborty, S., Detwiller, M., Kunduri, B. and McWilliams, K. (2022). pyDARN: A Python software for visualizing SuperDARN radar data. Front. Astron. Space Sci. 9:1022690. doi: 10.3389/fspas.2022.1022690

30. Chakraborty, S., Baker, J. B. H., and Ruohoniemi, J. M. (2021). Probabilistic Short-wave Fadeout Detection in SuperDARN Time Series Observations. 2021 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), pp. 43–48. doi: 10.1109/WiSEE50203.2021.9613835

31. Chakraborty, S., Qian, L., Ruohoniemi, J. M., Baker, J. B. H., Mclnerney, J. M., and Nishitani, N. (2021). The role of flare-driven ionospheric electron density changes on the Doppler flash observed by SuperDARN HF radars. Journal of Geophysical Research: Space Physics, 126, e2021JA029300. doi: 10.1029/2021JA029300

32. Fiori, R. A. D., Chakraborty, S., Nikitina, L. (2022). Data-based optimization of a simple shortwave fadeout absorption model. Journal of Atmospheric and Solar-Terrestrial Physics. doi: 10.1016/j.jastp.2022.105843

33. Chakraborty, S., Baker, J. B. H., Fiori, R. A. D., Ruohoniemi, J. M., and Zawdie, K. A. (2021). A modeling framework for estimating ionospheric HF absorption produced by solar flares. Radio Science, 56, e2021RS007285. doi:10.1029/2021RS007285

34. Chakraborty, S., Ruohoniemi, J. M., Baker, J. B. H., Fiori, R. A. D., Bailey, S. M., and Zawdie, K. A. (2021). Ionospheric Sluggishness: A Characteristic Time-Lag of the Ionospheric Response to Solar Flares. Journal of Geophysical Research: Space Physics, 126, e2020JA028813. doi: 10.1029/2020JA028813

35. Chakraborty, S. and Morley, S. K. (2020). Probabilistic Prediction of Geomagnetic Storms and the Kp Index. Journal of Space Weather and Space Climate. doi:10.1051/swsc/2020037

36. Mukhopadhyay, A., Welling, D. T., Liemohn, M. W., Ridley, A. J., Chakraborty, S., and Anderson, B. J. (2020). Conductance Model for Extreme Events: Impact of auroral conductance on space weather forecasts. Space Weather, 18, e2020SW002551. doi:10.1029/2020SW002551

37. Chakraborty, S., Baker, J. B. H., Ruohoniemi, J. M., Kunduri, B., Nishitani, N., and Shepherd, S. G. (2019). A Study of SuperDARN Response to Co-occurring Space Weather Phenomena. Space Weather, 17. doi:10.1029/2019SW002179

38. Chakraborty, S., Ruohoniemi, J. M., Baker, J. B. H., and Nishitani, N. (2018). Characterization of short-wave fadeout seen in daytime SuperDARN ground scatter observations. Radio Science, 53, 472–484. doi:10.1002/2017RS006488

39. Fiori, R. A. D., Koustov, A. V., Chakraborty, S., Ruohoniemi, J. M., Danskin, D. W., Boteler, D. H., and Shepherd, S. G. (2018). Examining the Potential of the Super Dual Auroral Radar Network for Monitoring the Space Weather Impact of Solar X-Ray Flares. Space Weather, 16. doi:10.1029/2018SW001905

CONFERENCE PAPERS

1. Chakraborty, S., Ruohoniemi, J. M., Baker, J. B. H., Fiori, R., Zawdie, K., Bailey, S., Nishitani, N., and Drob, D. et al. (2020). Sluggishness of the Ionosphere: Characteristic time-lag in Response to Solar Flares. 2020 XXXIIIrd General Assembly and Scientific Symposium of the International Union of Radio Science, Rome, Italy, pp. 1–4. doi: 10.23919/URSIGASS49373.2020.9232206

2. Chakraborty, S., Mukherjee, U. (2015). Circular Micro-strip Antenna Modelling using FDTD method and Design using Genetic Algorithms. CALCON-2015, ID: EC20140700015

3. Chakraborty, S., Mukherjee, U. (2014). Circular Micro-strip (Coax Feed) Antenna Modelling using FDTD Method and Design using Genetic Algorithms: A Comparative Study on Different Types of Design Techniques. ICCCT-2014, IEEE Digital Xplore. doi:10.1109/ICCCT.2014.7001514

4. Chakraborty, S., Mukherjee, U. (2011). Comparative Study of Micro-strip antennas designed by coaxial feed and line feed. ICCCT-2011, IEEE Digital Xplore. doi:10.1109/ICCCT.2011.6075196

5. Chakraborty, S., Mukherjee, U. (2010). Micro-strip antenna optimization using genetic algorithms. ICCCT-2010, IEEE Digital Xplore. doi:10.1109/ICCCT.2011.6075196

INVITED PRESENTATIONS

1. Chakraborty, S. (Invited). Characterization of Solar Flare Effects Observed by High Frequency Radar. HAO Colloquium, 12 July 2023, Online.
2. Chakraborty, S. (Invited), Shi, X., Chisham, G., Ruohoniemi, J. M., Baker, J. B. H., Stern, K., and Thomas, E. G. Coordinated Investigation of Antarctic Total Solar Eclipse (TSE) using SuperDARN HF Radars. CEDAR Workshop, Austin, June 2022.

3. Maimaiti, M., Chakraborty, S. (Invited). SuperDARN Radars in Space Science Research. CEDAR Workshop, Keystone, June 2017.

SELECTED PRESENTATIONS

1. Chakraborty, S., Nishitani, N., Ruohoniemi, J. M., Shi, X., Baker, J. B. H., & Ponomarenko, P. V. (2023, December). Solar flare-induced gradient drift instability observed by SuperDARN HF radars. AGU Fall Meeting Abstracts, SA31B-2861.
2. Shi, X., Chakraborty, S., Baker, J. B. H., Hartinger, M., Lin, D., Wang, W., Sterne, K. T. (2023, December). Quantifying Energy Deposition through Joule Heating from Ionospheric Ultra-Low-Frequency Perturbations using SuperDARN Data. AGU Fall Meeting Abstracts, SA34A-07.
3. Chakraborty, S., Nishitani, N., Ruohoniemi, J. M., Baker, J. B. H., Shi, X., Kunduri, B. Solar Flare effects on High Latitude Electrodynamics. Fall AGU, Chicago, IL, 12–16 December 2022.
4. Boteler, D., Chakraborty, S., Shi, X., Murphy, B., Hartinger, M., Wang, X., Lucas, G., and Baker, J. Modeling Geomagnetic Induction in Submarine Cable. Fall AGU, Chicago, IL, 12–16 December 2022.
5. Chakraborty, S., Nishitani, N., Qian, L., Ruohoniemi, J., Baker, J., Mclnerney, J. The Role of Flare-Driven Ionospheric Electron Density Changes on the Doppler Flash Observed by SuperDARN HF Radars. STE, Nagoya University, Japan, 27 September 2022.
6. Chakraborty, S., Qian, L., Mabie, J., Mclnerney, J. M., Mark, S., Erickson, P. J., Earle, G. Origination of Ionospheric G-condition following a Total solar eclipse. TESS, Bellevue, WA, 8–11 August 2022.
7. Chakraborty, S., Nishitani, N., Ruohoniemi, J. M., and Baker, J. B. H. Ionospheric Response to Solar Flares Observed in SuperDARN HF Radars. SoSpIM, Online, 27–28 June 2022.
8. Chakraborty, S., Qian, L., Ruohoniemi, J. M., Baker, J., and McInerney, J. The effects of solar flare-driven ionospheric electron density change on Doppler Flash. SuperDARN Workshop, 2021.
9. Chakraborty, S., Baker, J. B. H., Fiori, R. A. D., Ruohoniemi, J. M., & Zawdie, K. A. Testing the role of Dispersion Relation & Collision Frequency Formulations on Estimation of Shortwave-Fadeout (SWF). URSI GASS, Rome, Italy, 2021.
10. Chakraborty, S., Baker, J. B. H., Ruohoniemi, J. M., Bailey, S., Fiori, R. A. D., Nishitani, N. A Study of Solar Flare Effects on Mid and High Latitude Radio Wave Propagation using SuperDARN HF Radar. AGU Fall Meeting, Washington DC, 2018.

OUTREACH & ACTIVITIES

Participated in Summer Camp for pre-college students, including Imagination (rising 7th and 8th graders) and Pathways for Future Engineers (first-generation rising 10–12th graders), as a volunteer faculty member, organized by the Center for the Enhancement of Engineering Diversity (CEED) at Virginia Tech (since 2021).

Participated in Science and Collaborative talks for freshmen as a volunteer faculty member, organized by the Center for the Enhancement of Engineering Diversity (CEED) at Virginia Tech (since 2021).

INDUSTRIAL WORK EXPERIENCE

4.75 years of work experience at Tata Consultancy Services Innovation Labs as an IT Analyst. Worked on several projects based on a Home Energy Management System and a Device Management System, and on software development modules for a Platform-as-a-Service (PaaS) — including API and app gateways, an automated user-application deployment module, a distributed task queue, and a cloud-based sensor data explorer. Tools and software used: Python, Java, C, JavaScript, Lua, HTML, XML, JSON, CSS, Scala, Akka (parallel computation framework), Redis, and MySQL.

Also contributed significantly to the development of the new Virginia Tech SuperDARN website using the Python-Flask framework.

REFERENCES

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